

Class 14: Strength of weak ties and finding a job

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Sociology 204: Social Networks
Princeton University

Wednesday, March 19, 2025



New laptop policy

- ▶ Default: No laptops in lecture
- ▶ If you would like to use a laptop in lecture, please email me explaining your reasons so that we can learn from them. All reasonable requests will be accepted.
- ▶ All students using laptops must sit in the last row so as not to distract their classmates (again email me if you need a change).
- ▶ Questions?

More information:

<https://www.nytimes.com/2017/11/22/business/laptops-not-during-lecture-or-meeting.html>

<https://socialscience.msu.edu/news-events/news/archives/2021/2021-06-01.html>

<https://mcgraw.princeton.edu/guiding-students-using-technology-appropriately-suggestions-your-syllabus>

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- ▶ changes and stability of core discussion networks of Americans from 1985 - 2022, including during COVID

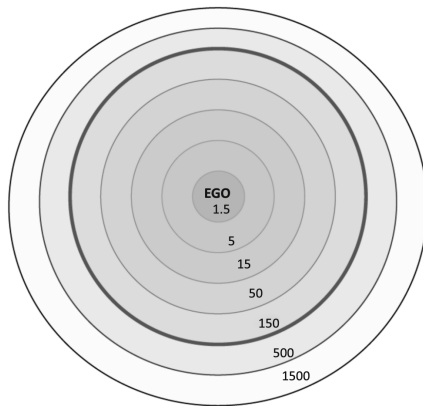
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- ▶ changes and stability of core discussion networks of Americans from 1985 - 2022, including during COVID
- ▶ the size of core discussion networks is shrinking, but doesn't seem to have been impacted by COVID
- ▶ COVID seems to have lead to an increase in homophily by partisanship in core discussion networks

1. Granovetter, M. (1973). The strength of weak ties. *American Journal of Sociology*, 78(6), 1360-1380.
2. Wang and Uzzi (2022). Weak ties, failed tries, and success. *Science*, 377(6612), 1256-1258.
3. Rajkumar et al. (2022). A causal test of the strength of weak ties. *Science*, 377(6612), 1304-1310.
4. Singer (2022) LinkedIn Ran Social Experiments on 20 Million Users Over Five Years. *New York Times*.

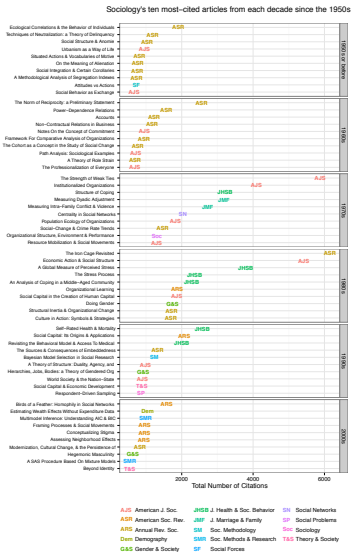


Trends in Cognitive Sciences

Figure 2. The Circles of Friendship. An individual (ego) sits within a series of hierarchically inclusive circles, or layers, of friendship that have a very distinct scaling ratio (each layer is three times the size of the one inside it). The heavy line demarcates the 150 layer, the typical size of personal social networks. Beyond this are acquaintances (500 layer); the layer at 1500 appears to represent the limit on the number of faces we can put names to. Frequency of contact, rated emotional closeness, and willingness to act altruistically towards a given alter all decline across the successive circles [2,10].

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- ▶ one extra bit of complexity and one plausible assumption lead to surprising empirical predictions, many that turn out to be true
- ▶ connects micro rules to macro patterns (similar to Bearman et al. study of sexual networks)
- ▶ generates lots of new research

AMERICAN SOCIOLOGICAL REVIEW

Karl F. Schuessler, Editor

Marvin E. Olsen, Book Review Editor

Sycamore Hall
Indiana University
Bloomington, Indiana 47401
Telephone 812/337-2385

December 9, 1969

Mr. Mark S. Granovetter
Lowell House M-21
Cambridge, Massachusetts 02138

Dear Mr. Granovetter:

Thank you for the opportunity to consider your article entitled "Alienation Reconsidered: The Strength of Weak Ties" for possible publication in ASR. We have completed our evaluation and reluctantly regret to notify you of our decision against publication. The reasons for our decision are set forth in suggestions prepared by our readers for your use. Perhaps these quite generous comments will be of some service to you should you resubmit your materials before submitting your manuscript for publication elsewhere.

Again, our appreciation for the opportunity to examine your paper.

Sincerely yours,



Karl Schuessler
Editor

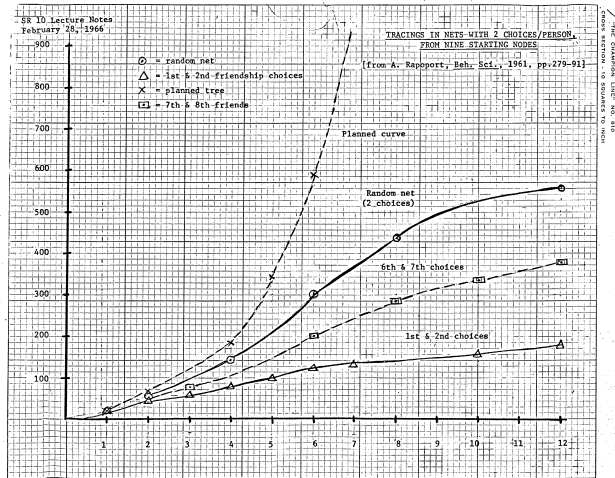
KS/ar

Enclosures: comments of two readers

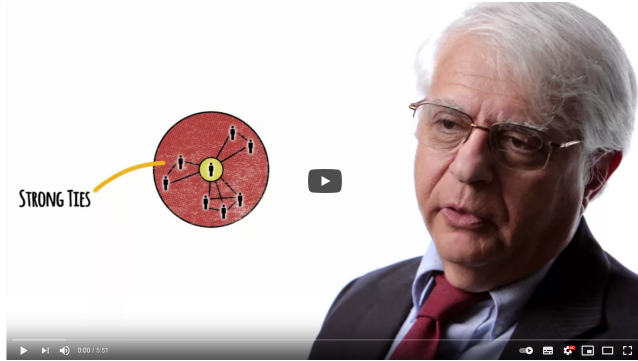
MS returned under separate cover

Official Journal of the American Sociological Association

<https://scatter.wordpress.com/2014/10/13/granovetter-rejection/>



Source: From Harrison White's class in 1966



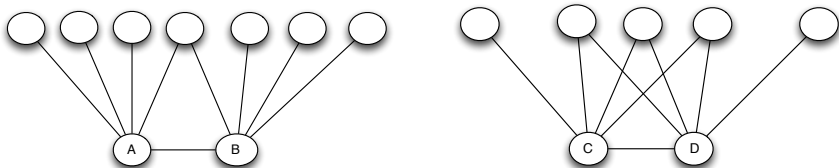
<https://www.youtube.com/watch?v=g3bBajcR5fE>

Let's listen to Granovetter's story behind the research and how it happened

“Strength of a tie is a (probably linear) combination of the amount of time, emotional intensity, the intimacy, and reciprocal services which characterize the tie.”

Prediction:

The stronger the tie between two people, the more their friendship sets overlap.



No good data to test in 1973. That's sometimes a sign of a truly new idea.

For the rest of the paper, he deals with a simplification:

- ▶ Old idea: tie is either present or absent
- ▶ New idea: tie is either strong, weak, or absent

Assume no forbidden triad

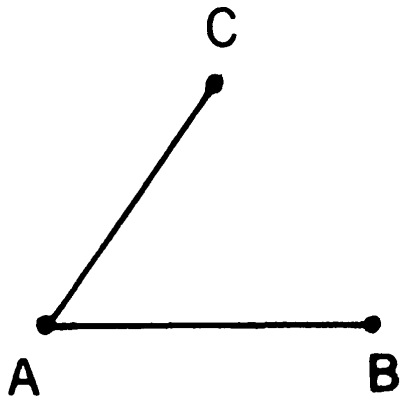


FIG. 1.—Forbidden triad

Except in weird cases, *no strong tie is a bridge*. Or *all bridges are weak ties*.

But, it is not the case that all weak ties are bridges.

Bridges are probably rare, so Granovetter talks about *local bridges*

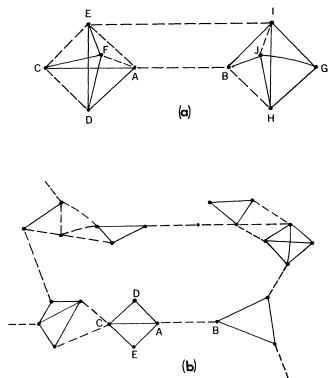


FIG. 2.—Local bridges. *a*, Degree 3; *b*, Degree 13. — = strong tie; --- = weak tie.

“The significance of weak ties, then, would be that those which are local bridges create more and shorter paths.”

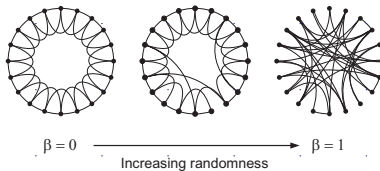
Which concept or model that we learned about does this remind you of?

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Shortcuts in beta model

3.6



Strength of weak ties for

- ▶ communities
- ▶ individuals

Strength of weak ties for communities

Boston's West End was unable to resist "urban renewal", Granovetter argues that this is because of lack of short paths between people (think "caveman graph")

Granovetter predicts: weak ties hold communities together and enable them to act cohesively

Strength of weak ties for individuals

Local bridges are most important for spreading new things. Therefore, weak ties are important for spreading new things and receiving information.

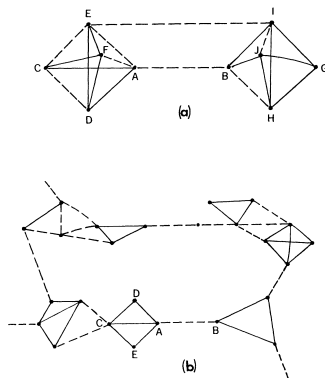


FIG. 2.—Local bridges. *a*, Degree 3; *b*, Degree 13. — = strong tie; - - - = weak tie.

In the sample he studied (professionals in a suburb of Boston), of those who found a job through a contact, how often did they see the contact?

- ▶ often (at least twice a week): 17%
- ▶ occasionally (more than once a year but less than twice a week): 56%
- ▶ rarely (once a year or less): 28%

The people you see occasionally and rarely can be the most important.

¹⁵ Although this corresponds only to the first of the four dimensions in my definition, supplementary anecdotal evidence from interviews makes it likely that, in this case, the entire definition is satisfied by this measure. At the time of research, it had not occurred to me that tie strength would be a useful variable.

To summarize lots of subsequent research, it is non-redundant information that is important, not weak ties.

Therefore, a better title might have been: *The strength of local bridges*

¹⁸ Donald Light has suggested to me an alternative reason to expect predominance of weak ties in transfer of job information. He reasons that most of any given person's ties are weak, so that we should expect, on a "random" model, that most ties through which job information flows should be weak. Since baseline data on acquaintance networks are lacking, this objection remains inconclusive. Even if the premise were correct, however, one might still expect that greater motivation of close friends would overcome their being outnumbered. Different assumptions yield different "random" models; it is not clear which one should be accepted as a starting point. One plausible such model would expect information to flow through ties in proportion to the time expended in interaction; this model would predict much more information via strong ties than one which merely counted all ties equally.

Maybe people find jobs through weak ties because they have more weak ties?

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Weak Ties, Information, and Influence: How Workers Find Jobs in a Local Russian Labor Market

Valery Yakubovich
University of Chicago

<http://asr.sagepub.com/content/70/3/408.full.pdf+html>

Probability of finding a job through a weak tie is higher than finding a job through a strong tie (in Samara, Russia in 1998), but there are some caveats

Strength of weak ties for job search moves online, Facebook

Journal of Economic Behavior & Organization 133 (2017) 362–372



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Contents lists available at ScienceDirect

Journal of Economic Behavior & Organization

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The paradox of weak ties in 55 countries



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James H. Fowler^{e,f}

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^b Department of Sociology, SUNY Stony Brook, United States

^c Department of Political Science, Pennsylvania State University, United States

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^f Department of Political Science, University of California, San Diego, United States

“In our US study we identify a seeming paradox of weak ties. The paradox is that most people are helped through one of their numerous weak ties, but a single stronger tie is significantly more valuable at the margin.”

A causal test of the strength of weak ties

Karthik Rajkumar¹, Guillaume Saint-Jacques¹, Iavor Bojinov², Erik Brynjolfsson^{3,4}, Sinan Aral^{5*}

Consider a dream experiment, where the prevalence of strong and weak ties within social networks was randomly varied to see whether these variations cause any changes in the probability of finding a job.

Source: Wang and Uzzi (2022)

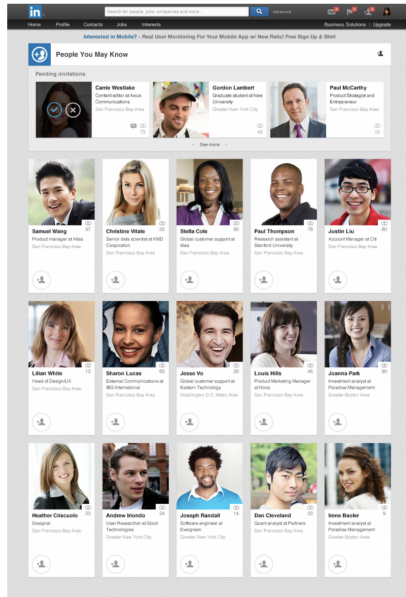


Figure S1: A sample My Network Page on LinkedIn.com showing the recommendations generated by the People You May Know algorithm.

Ensemble machine learning model combining:

- ▶ estimated propensity that ego will send an invite to alter
- ▶ estimated propensity that alter will accept an invite from ego
- ▶ estimated engagement between ego and alter once connected
- ▶ weights of each of these models

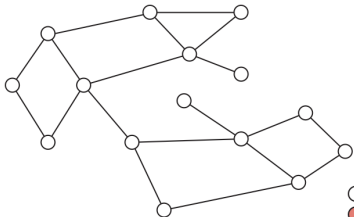
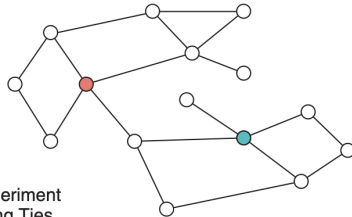
This stuff is the life of data scientists (“the sexiest jobs of the 21st century”) at tech companies.

Measurement of tie strength

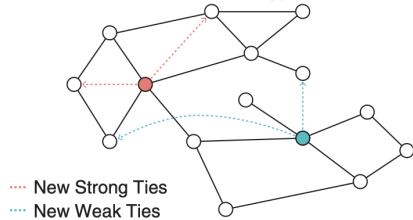
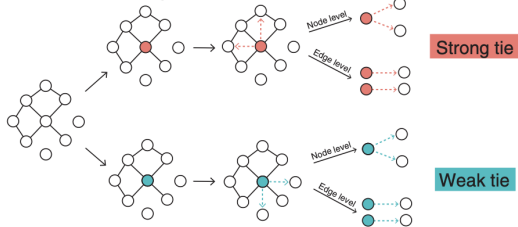
We measured tie strength by its two leading indicators: the intensity of the interaction between two people and the number of mutual connections they had in common. We measured

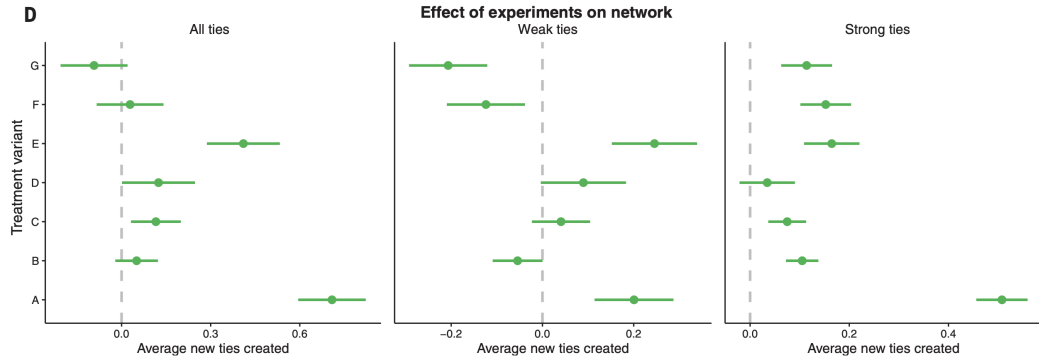
Measurement of sequential job

an experiment. In accordance with the literature (25, 26), we consider a job transmission to have occurred when three criteria are satisfied: First, user A reports working at company c at date D_1 . Second, user B reports working at that same company c at a later date D_2 , with D_2 and D_1 being at least one year apart. Third, user A and user B were friends on the social network at least one full year before D_2 . In the weak tie literature, when these three criteria are met, a tie is considered a “sequential job” tie, which represents the state of the art in measuring relational job mobility.

A**Pre-treatment network****Intervention**

- Did Not Trigger Into Experiment
- Recommend More Strong Ties
- Recommend More Weak Ties

Creation of different types of ties**Representation of data**



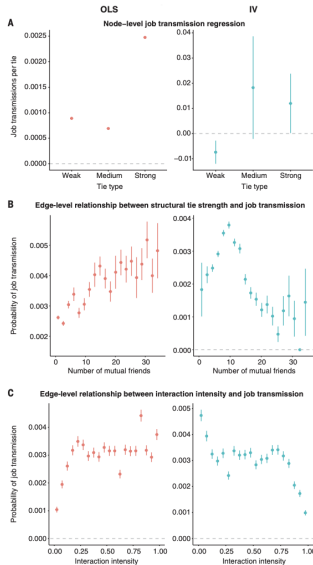


Fig. 3. The causal effect of tie strength on job mobility. The figure displays the estimated effects of tie strength on job transmissions at the node and edge levels. In each panel the left column displays the results of the OLS analysis whereas the right column displays the experimental IV results from first wave experiments conducted in 2015. **(A)** displays the effect of weak, medium, and strong ties (defined by terciles of mutual friends) on job transmissions aggregated to the node level; **(B)** displays the effects of structural tie strength based on the number of mutual friends on job transmissions at the edge level whereas **(C)** displays the effects of tie strength based on interaction intensity on job transmissions at the edge level.

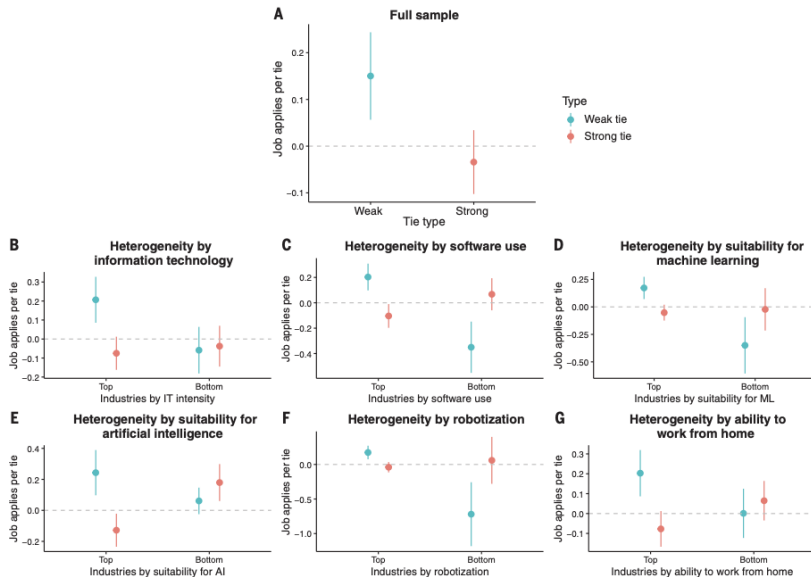


Fig. 4. Heterogeneous causal effects by industry. (A) displays experimental IV analysis of the effects of weak and strong ties on job applications at the node level in the full sample. The rest of the panels display experimental IV analysis of the heterogeneous effects of weak and strong ties on job applications

at the node level to jobs in industries in the top and bottom half of the industry distributions of (B) IT intensity; (C) software intensity; (D) suitability for machine learning; (E) suitability for artificial intelligence; (F) degree of robotization; and (G) suitability for remote work.

The authors analyzed data from multiple large-scale randomized experiments on LinkedIn's People You May Know algorithm, which recommends new connections to LinkedIn members, to test the extent to which weak ties increased job mobility in the world's largest professional social network. The experiments randomly varied the prevalence of weak ties in the networks of over 20 million people over a 5-year period, during which 2 billion new ties and 600,000 new jobs were created. The results provided experimental causal evidence supporting the strength of weak ties and suggested three revisions to the theory. First, the strength of weak ties was nonlinear. Statistical analysis found an inverted U-shaped relationship between tie strength and job transmission such that weaker ties increased job transmission but only to a point, after which there were diminishing marginal returns to tie weakness. Second, weak ties measured by interaction intensity and the number of mutual connections displayed varying effects. Moderately weak ties (measured by mutual connections) and the weakest ties (measured by interaction intensity) created the most job mobility. Third, the strength of weak ties varied by industry. Whereas weak ties increased job mobility in more digital industries, strong ties increased job mobility in less digital industries.

Notice how the abstract tell you what they did (yellow) and what they found (blue)

Which evidence do you find more convincing about the importance of weak ties in job search?

1. Granovetter (1973)
2. Rajkumar et al. (2022)

LinkedIn Ran Social Experiments on 20 Million Users Over Five Years

A study that looked back at those tests found that relatively weak social connections were more helpful in finding jobs than stronger social ties.

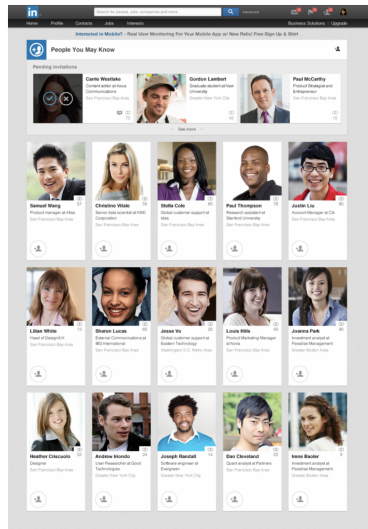


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Three principles from the Belmont Report

- ▶ Autonomy
- ▶ Beneficence
- ▶ Justice

Provided reasonable guidance for analogue era research, trickier in digital age

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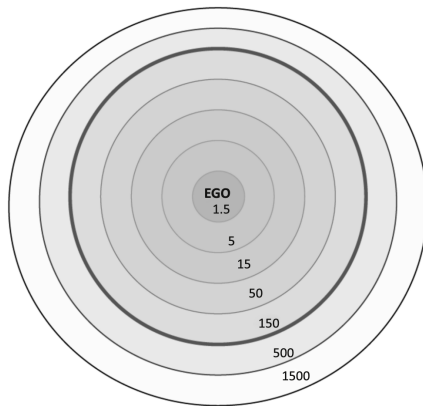
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- ▶ weak ties also help hold communities together
- ▶ the relationship between tie strength and probability of finding a job is non-linear and depends on exactly how you measure tie strength



Trends in Cognitive Sciences

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- ▶ Reeves and Fall (2023) Seven key takeaways from Chetty's new research on friendship and economic mobility Brookings Institution.
- ▶ Angrist and Sacerdote (2022) The social connections that shape economic prospects. Nature 608, 37-38.
- ▶ Chetty et al. (2022) Social capital I: measurement and associations with economic mobility. Nature 608, 108-121.
- ▶ Chetty et al. (2022) Social capital II: determinants of economic connectedness. Nature 608, 122-134.